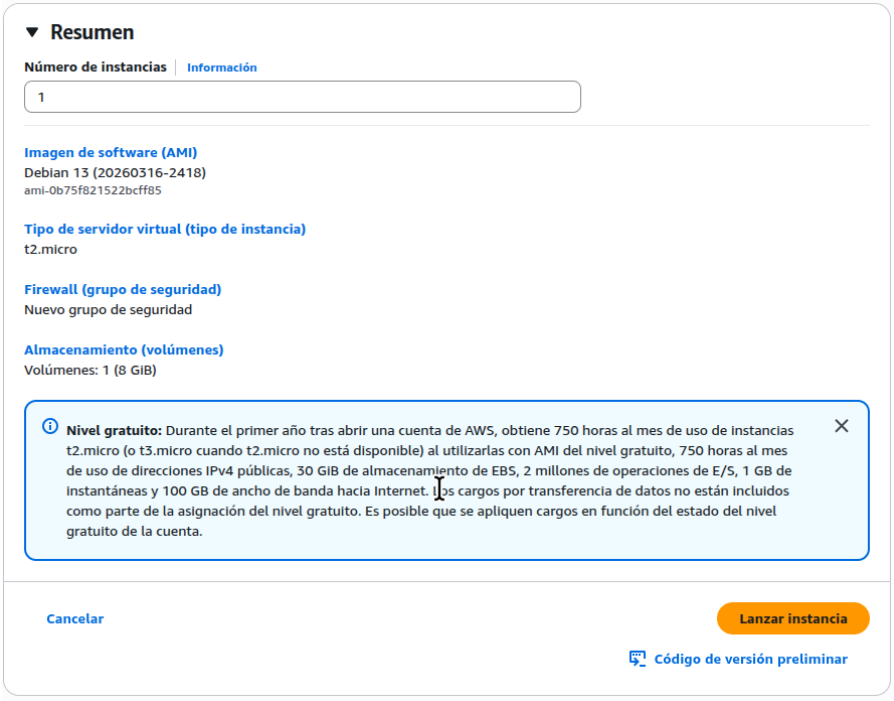
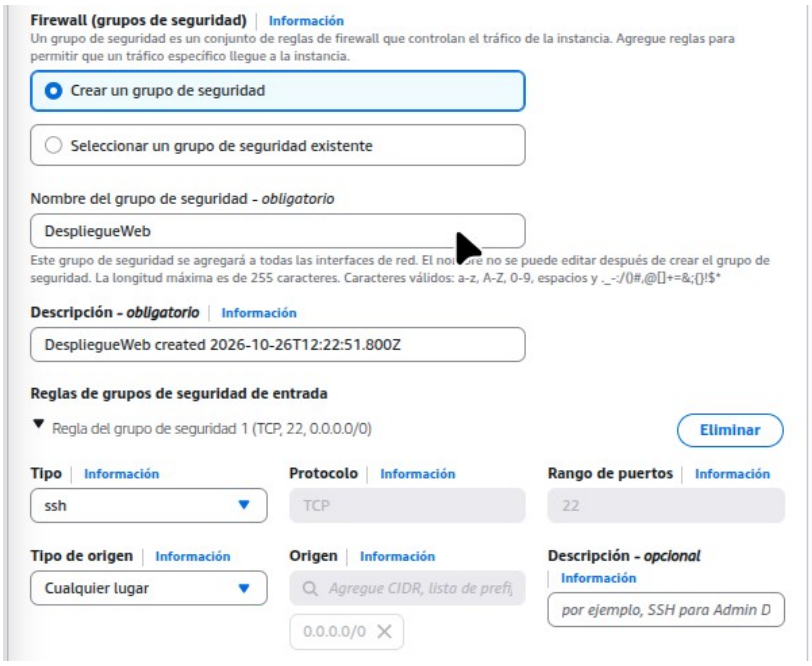
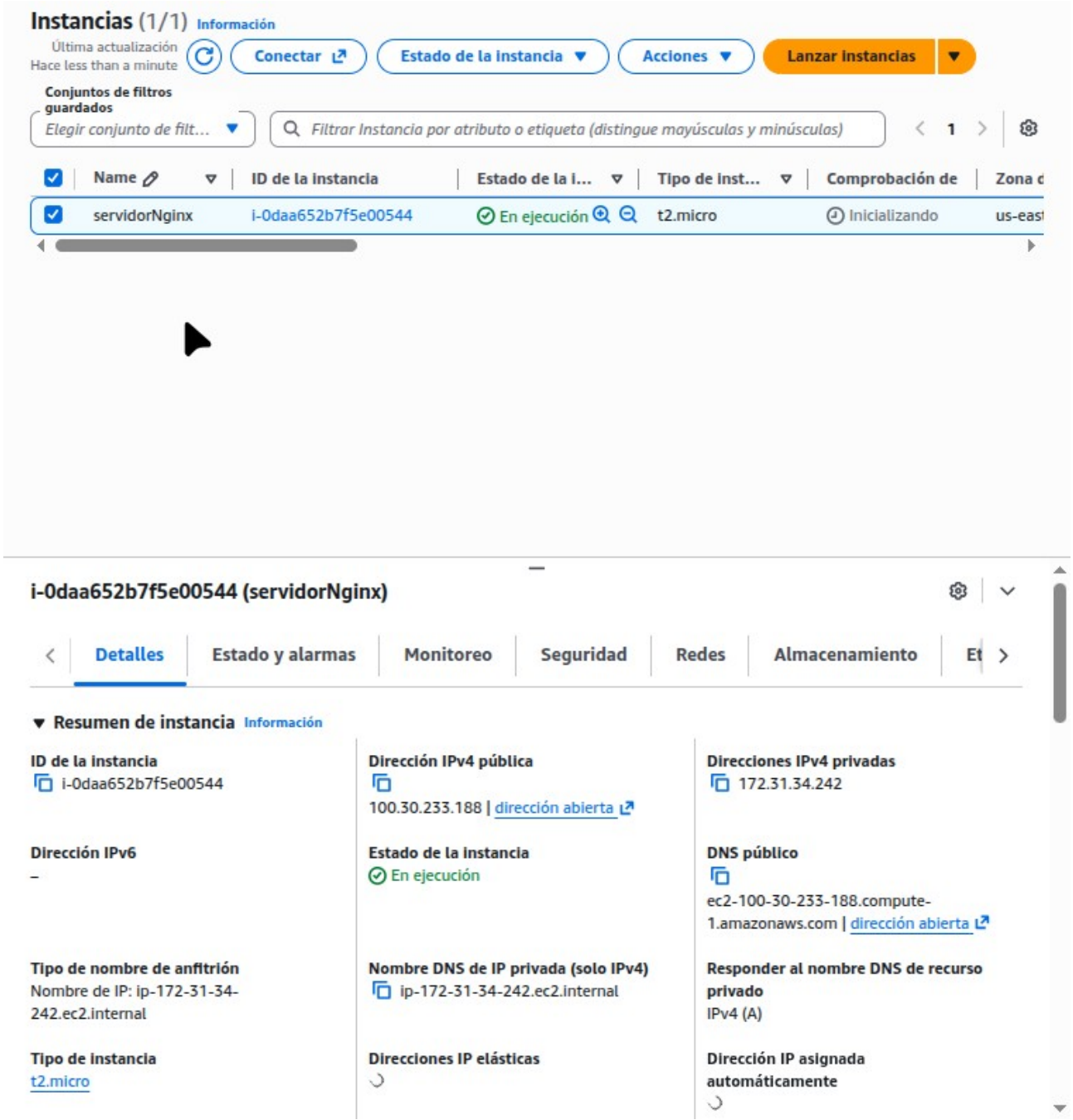
# Nginx AWS

Se va a crear una instancia en AWS con estas caracteristicas:

- Nombre: servidorNginx
- SO: Debian
- Tipo de instancia: t2.micro
- Par de clave: DespliegueDAW
- Grupo de seguridad:
  - DespliegueWeb con acceso por SSH solo






Nos conectamos por ssh cambiando los permisos de la clave .pem para que solo el usuario dueño tenga permisos de lectura

```
ivan@ivan-kubuntu ~/Documentos/2º DAW/Despliege aplicaiones Web
❯ chmod 400 "DespliegueDAW.pem"
ivan@ivan-kubuntu ~/Documentos/2º DAW/Despliege aplicaiones Web
❯ ssh -i "DespliegueDAW.pem" admin@ec2-100-30-233-188.compute-1.amazonaws.com
ksshaskpass: Unable to parse phrase "The authenticity of host 'ec2-100-30-233-188.compute-1.amazonaws.com (100.30.23
3.188)' can't be established.\nED25519 key fingerprint is: SHA256:ZI4opLy9Sd7R8uKL/g137RBY12ad0CpwRqImvfpHOnk\nThis
key is not known by any other names.\nAre you sure you want to continue connecting (yes/no/[fingerprint])? "
Warning: Permanently added 'ec2-100-30-233-188.compute-1.amazonaws.com' (ED25519) to the list of known hosts.
Linux ip-172-31-34-242 6.12.74+deb13+1-cloud-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.12.74-2 (2026-03-08) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
admin@ip-172-31-34-242:~$
```

Se actualizan los paquetes y se instala el paquete nginx

```
admin@ip-172-31-34-242:~$ sudo apt update && sudo apt upgrade
Get:1 file:/etc/apt/mirrors/debian.list Mirrorlist [38 B]
Get:5 file:/etc/apt/mirrors/debian-security.list Mirrorlist [47 B]
Get:2 https://cdn-aws.deb.debian.org/debian trixie InRelease [140 kB]
Get:3 https://cdn-aws.deb.debian.org/debian trixie-updates InRelease [47.3 kB]
Get:4 https://cdn-aws.deb.debian.org/debian trixie-backports InRelease [54.0 kB]
Get:6 https://cdn-aws.deb.debian.org/debian-security trixie-security InRelease [43.4 kB]
Get:7 https://cdn-aws.deb.debian.org/debian trixie/main Sources [10.5 MB]
Get:8 https://cdn-aws.deb.debian.org/debian trixie/main amd64 Packages [9678 kB]
Get:9 https://cdn-aws.deb.debian.org/debian trixie/main Translation-en [6486 kB]
Get:10 https://cdn-aws.deb.debian.org/debian trixie-updates/main Sources [820 B]
Get:11 https://cdn-aws.deb.debian.org/debian trixie-updates/main amd64 Packages [720 B]
Get:12 https://cdn-aws.deb.debian.org/debian trixie-updates/main Translation-en [772 B]
Get:13 https://cdn-aws.deb.debian.org/debian trixie-backports/main Sources [333 kB]
Get:14 https://cdn-aws.deb.debian.org/debian trixie-backports/main amd64 Packages [294 kB]
Get:15 https://cdn-aws.deb.debian.org/debian trixie-backports/main Translation-en [230 kB]
Get:16 https://cdn-aws.deb.debian.org/debian-security trixie-security/main Sources [241 kB]
Get:17 https://cdn-aws.deb.debian.org/debian-security trixie-security/main amd64 Packages [270 kB]
Get:18 https://cdn-aws.deb.debian.org/debian-security trixie-security/main Translation-en [166 kB]
89% [7 Sources store 0 B]
```

```
admin@ip-172-31-34-242:~$ sudo apt install nginx
Installing:
  nginx

Installing dependencies:
  nginx-common

Suggested packages:
  fcgiwrap  nginx-doc  ssl-cert

Summary:
  Upgrading: 0, Installing: 2, Removing: 0, Not Upgrading: 0
  Download size: 725 kB
  Space needed: 1906 kB / 6621 MB available

Continue? [Y/n]
```

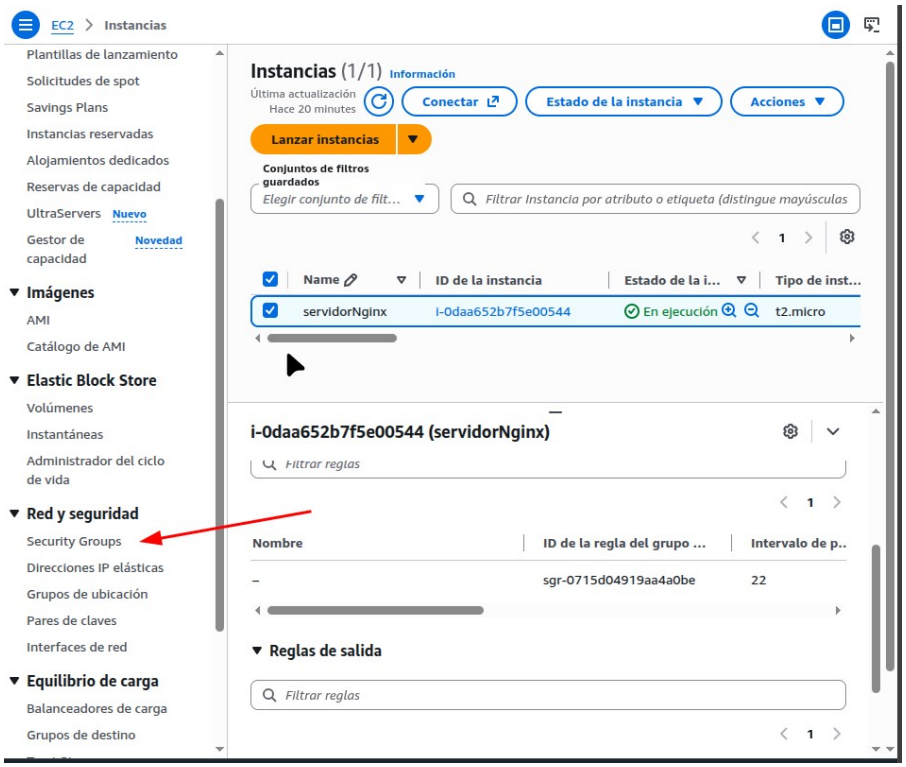
Una vez instalado se comprueba que el servicio de nginx este corriendo

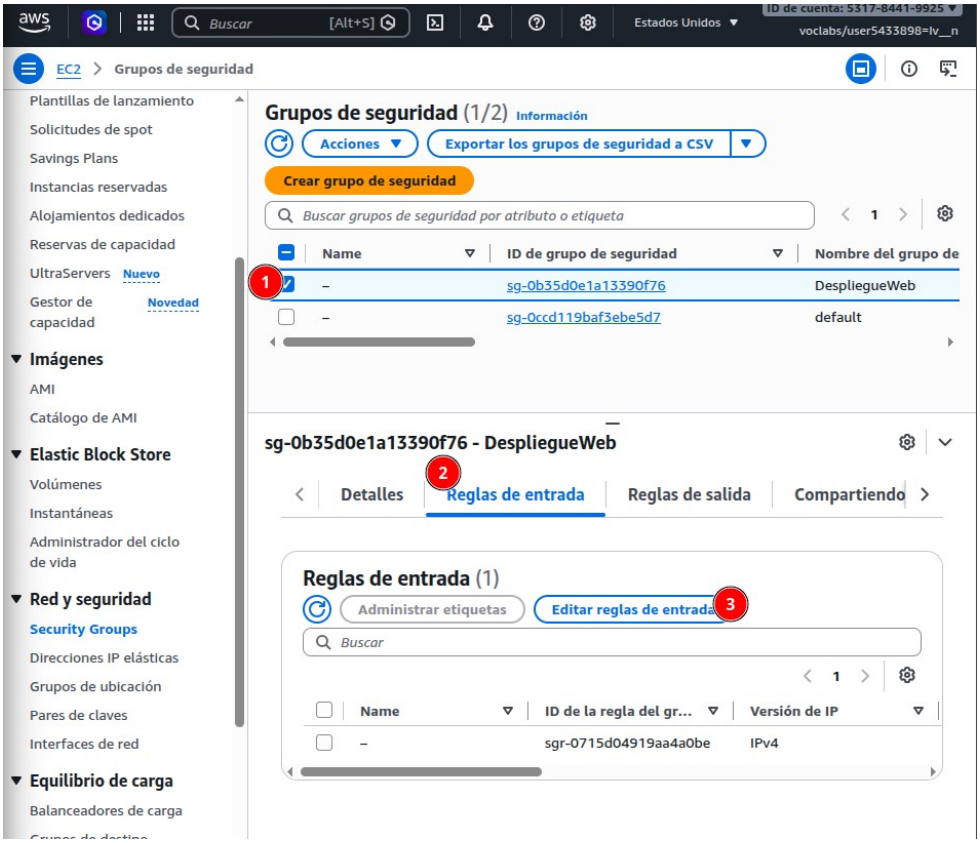
```
admin@ip-172-31-34-242:~$ systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
     Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
     Active: active (running) since Mon 2026-10-26 12:38:20 UTC; 1min 3s ago
 Invocation: eede5a85f1a24ca6a1201915f347a905
       Docs: man:nginx(8)
    Process: 15903 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on; master_process on; (code=exited, status=0/SUCCE>
    Process: 15905 ExecStart=/usr/sbin/nginx -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
   Main PID: 15931 (nginx)
      Tasks: 2 (limit: 1127)
     Memory: 2.2M (peak: 4.9M)
        CPU: 31ms
     CGroup: /system.slice/nginx.service
             ├─15931 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
             └─15934 "nginx: worker process"

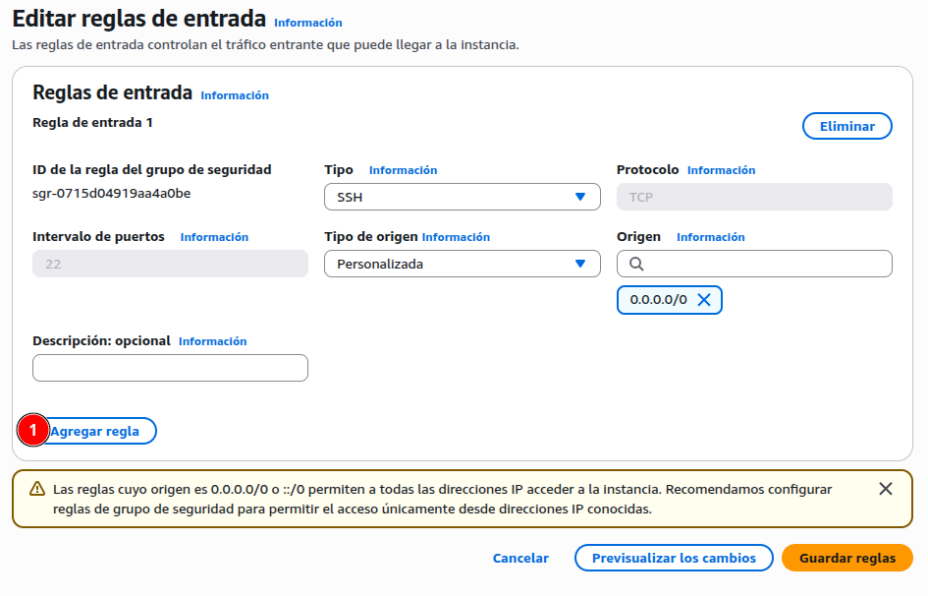
Oct 26 12:38:20 ip-172-31-34-242 systemd[1]: Starting nginx.service - A high performance web server and a reverse p>
Oct 26 12:38:20 ip-172-31-34-242 systemd[1]: Started nginx.service - A high performance web server and a reverse pr>
lines 1-17/17 (END)
```

Al intentar conectarse no llega nunca a la pagina web ya que falta configurar el grupo de seguridad (firewall) para permitir las conexiones entrantes a través del puerto 80 (HTTP).

Se accede a los grupos de seguridad y en el creado y vinculado con la instancia y se añade una regla de entrada con el tipo HTTP y el tipo de origen desde cualquier IPv4 y se guardan las reglas de entrada.

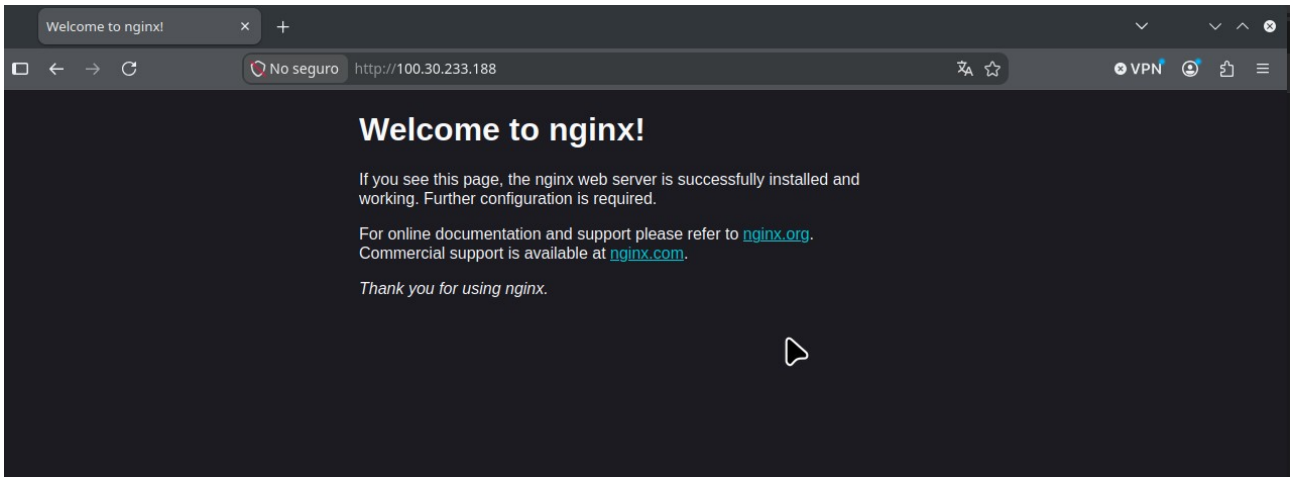



Buscar [Alt+S]
Estados Unidos
ID de cuenta: 5317-8441-9925
voclabs/user5433898=lv__n
EC2 > Grupos de seguridad
Plantillas de lanzamiento
Solicitudes de spot
Savings Plans
Instancias reservadas
Alojamientos dedicados
Reservas de capacidad
UltraServers Nuevo
Gestor de capacidad Novedad
Imágenes
AMI
Catálogo de AMI
Elastic Block Store
Volúmenes
Instantáneas
Administrador del ciclo de vida
Red y seguridad
Security Groups
Direcciones IP elásticas
Grupos de ubicación
Pares de claves
Interfaces de red
Equilibrio de carga
Balanceadores de carga
Grupos de seguridad (1/2) Información
Acciones
Exportar los grupos de seguridad a CSV
Crear grupo de seguridad
Buscar grupos de seguridad por atributo o etiqueta
Name
ID de grupo de seguridad
Nombre del grupo de
sg-0b35d0e1a13390f76
DespliegueWeb
sg-0ccd119baf3ebe5d7
default
1
sg-0b35d0e1a13390f76 - DespliegueWeb
2
Detalles
Reglas de entrada
Reglas de salida
Compartiendo
Reglas de entrada (1)
Administrar etiquetas
Editar reglas de entrada
3
Buscar
Name
ID de la regla del gr...
Versión de IP
sgr-0715d04919aa4a0be
IPv4

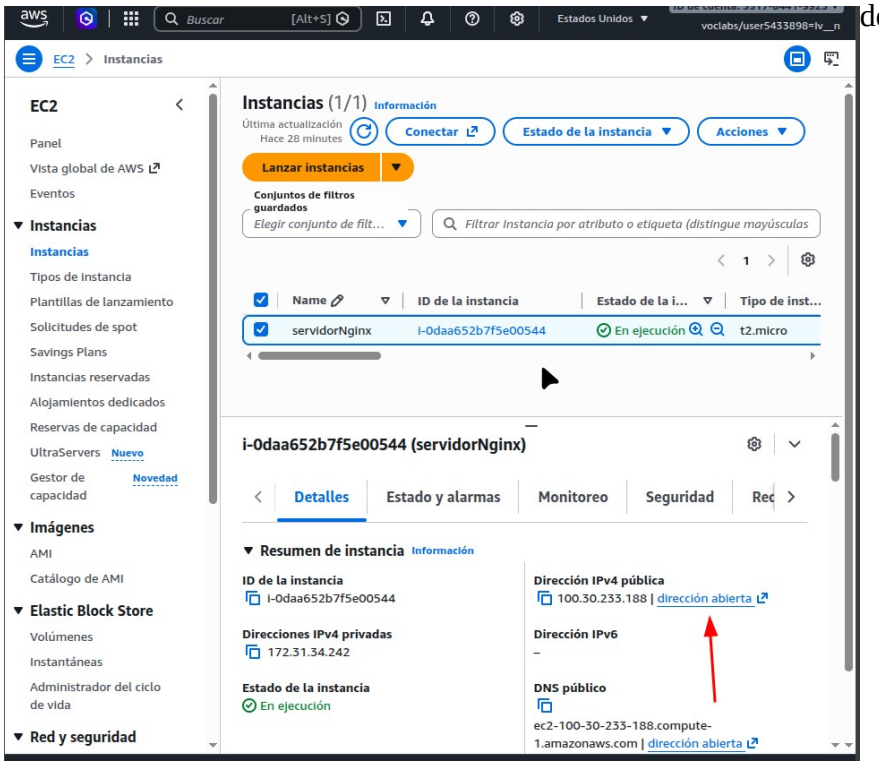

Editar reglas de entrada Información
Las reglas de entrada controlan el tráfico entrante que puede llegar a la instancia.
Reglas de entrada Información
Regla de entrada 1
Eliminar
ID de la regla del grupo de seguridad
sgr-0715d04919aa4a0be
Tipo Información
SSH
Protocolo Información
TCP
Intervalo de puertos Información
22
Tipo de origen Información
Personalizada
Origen Información
0.0.0.0/0
Descripción: opcional Información
1
Agregar regla
Las reglas cuyo origen es 0.0.0.0/0 o ::/0 permiten a todas las direcciones IP acceder a la instancia. Recomendamos configurar reglas de grupo de seguridad para permitir el acceso únicamente desde direcciones IP conocidas.
Cancelar
Previsualizar los cambios
Guardar reglas

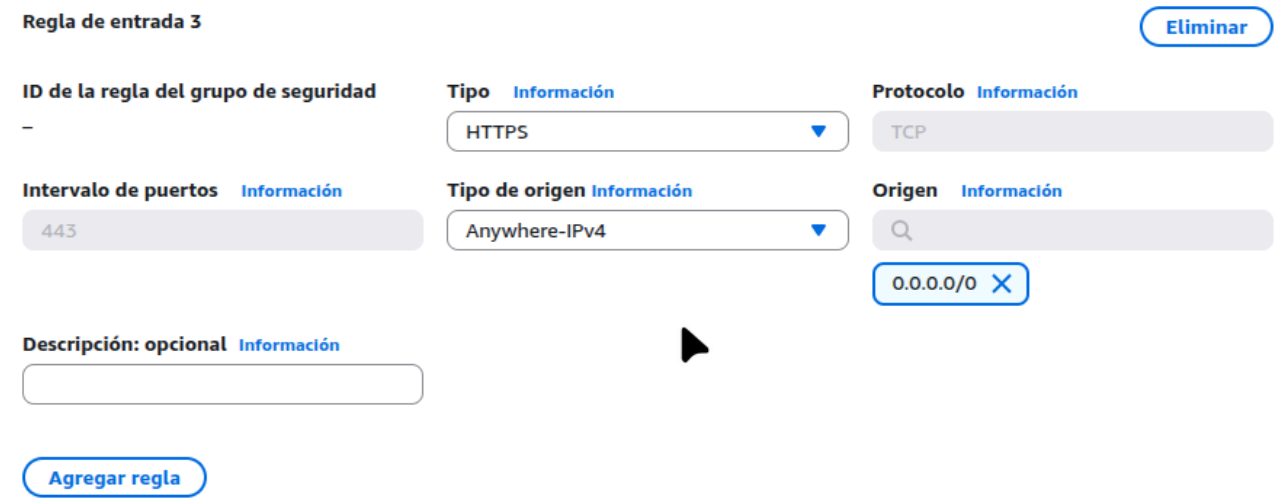



Se accede a la ip publica y se comprueba que se puede acceder




Ahora ya se puede acceder pero como la practica tambien pide abrir el puerto 443 (HTTPS) se repiten estos pasos para abrir ese puerto, aunque de momento no se usará porque nginx usa por defecto el puerto 80









Para comprobar que estamos usando la pagina de `/var/www/html/index.nginx-debian.html` vamos a editar el archivo y ver si se modifica en el navegador

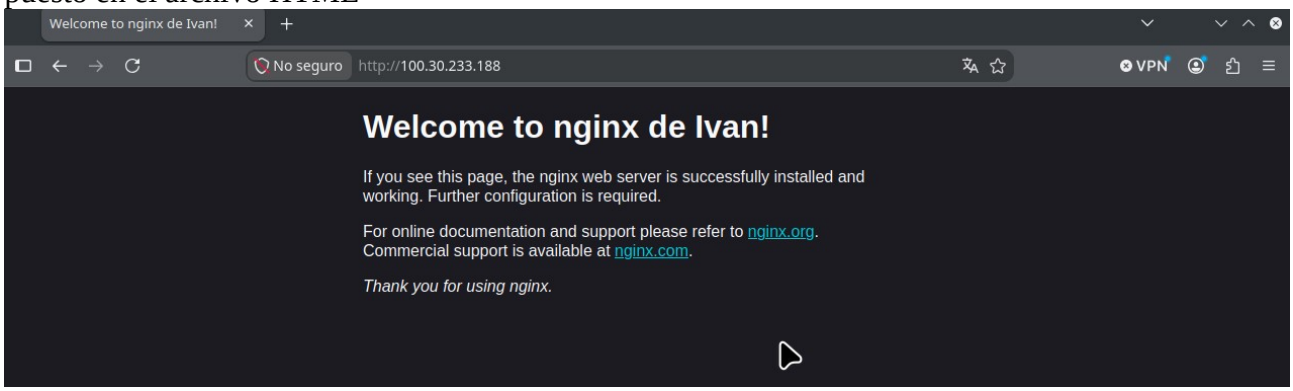
```
admin@ip-172-31-34-242:~$ sudo nano /var/www/html/index.nginx-debian.html
```

```
 GNU nano 8.4                                  /var/www/html/index.nginx-debian.html *
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx de Ivan!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx de Ivan!</h1>
<p>If you see this page, the nginx web server is successfully installed and
working. Further configuration is required.</p>

<p>For online documentation and support please refer to
<a href="http://nginx.org/">nginx.org</a>.<br/>
Commercial support is available at
<a href="http://nginx.com/">nginx.com</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
```

Volvemos a entrar en la pagina web en el navegador y vemos que ha cambiado como lo hemos puesto en el archivo HTML




Se comprueban los accesos correctos y erróneos con los logs del propio nginx

- access.log

```
admin@ip-172-31-34-242:~$ cat /var/log/nginx/access.log
207.175.10.158 - - [26/Oct/2026:13:13:25 +0000] "GET / HTTP/1.1" 200 409 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/127.0.0.0 Safari/537.36"
34.14.118.114 - - [26/Oct/2026:13:13:27 +0000] "\x16\x03\x00\x00i\x01\x00\x00e\x03\x03U\x1C\xA7\xE4random1random2random3random4\x00\x00\x0C\x00/\x00" 400 150 "-" "-"
34.14.118.114 - - [26/Oct/2026:13:13:27 +0000] "GET / HTTP/1.1" 200 615 "-" "Mozilla/5.0 (compatible; nmap-http-info)"
34.14.118.114 - - [26/Oct/2026:13:13:27 +0000] "OPTIONS / HTTP/1.1" 405 150 "-" "Mozilla/5.0 (compatible)"
34.14.118.114 - - [26/Oct/2026:13:13:27 +0000] "XSDX / HTTP/1.1" 405 150 "-" "Mozilla/5.0 (compatible)"
34.14.118.114 - - [26/Oct/2026:13:13:27 +0000] "GET /favicon.ico HTTP/1.1" 404 146 "-" "Mozilla/5.0 (compatible)"
34.14.118.114 - - [26/Oct/2026:13:13:27 +0000] "GET / HTTP/1.1" 200 615 "-" "Mozilla/5.0 (compatible)"
34.14.118.114 - - [26/Oct/2026:13:13:27 +0000] "GET /pub/ HTTP/1.1" 404 146 "-" "Mozilla/5.0 (compatible)"
34.14.118.114 - - [26/Oct/2026:13:13:27 +0000] "HEAD / HTTP/1.1" 200 0 "-" "Mozilla/5.0 (compatible)"
34.14.118.114 - - [26/Oct/2026:13:13:28 +0000] "POST / HTTP/1.1" 405 150 "-" "Mozilla/5.0 (compatible)"
34.14.118.114 - - [26/Oct/2026:13:13:28 +0000] "OPTIONS / HTTP/1.1" 405 150 "-" "Mozilla/5.0 (compatible)"
34.14.118.114 - - [26/Oct/2026:13:13:28 +0000] "GET /files/ HTTP/1.1" 404 146 "-" "Mozilla/5.0 (compatible)"
34.14.118.114 - - [26/Oct/2026:13:13:29 +0000] "GET /downloads/ HTTP/1.1" 404 146 "-" "Mozilla/5.0 (compatible)"
179.159.245.70 - - [26/Oct/2026:13:25:46 +0000] "GET / HTTP/1.1" 200 409 "-" "Mozilla/5.0"
20.38.35.133 - - [26/Oct/2026:14:06:46 +0000] "GET /developmentserver/metadatauploader HTTP/1.1" 404 118 "-" "Mozilla/5.0 zgrab/0.x"
142.93.84.144 - - [26/Oct/2026:14:42:10 +0000] "GET / HTTP/1.1" 200 409 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64; rv:156.0) Gecko/20100101 Firefox/156.0"
79.112.48.234 - - [26/Oct/2026:14:42:57 +0000] "GET / HTTP/1.1" 200 409 "-" "Mozilla/5.0 (X11; Ubuntu; Linux x86_64; rv:154.0) Gecko/20100101 Firefox/154.0"
79.112.48.234 - - [26/Oct/2026:14:42:57 +0000] "GET /favicon.ico HTTP/1.1" 404 118 "http://100.30.233.188/" "Mozilla/5.0 (X11; Ubuntu; Linux x86_64; rv:154.0) Gecko/20100101 Firefox/154.0"
167.172.59.78 - - [26/Oct/2026:14:43:34 +0000] "GET / HTTP/1.1" 200 409 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64; rv:156.0) Gecko/20100101 Firefox/156.0"
79.112.48.234 - - [26/Oct/2026:14:49:36 +0000] "GET / HTTP/1.1" 200 409 "-" "Mozilla/5.0 (X11; Linux x86_64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/156.0.0.0 Safari/537.36"
79.112.48.234 - - [26/Oct/2026:14:49:36 +0000] "GET /favicon.ico HTTP/1.1" 404 181 "http://100.30.233.188/" "Mozilla/5.0 (X11; Linux x86_64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/156.0.0.0 Safari/537.36"
79.112.48.234 - - [26/Oct/2026:14:56:48 +0000] "GET / HTTP/1.1" 200 413 "-" "Mozilla/5.0 (X11; Ubuntu; Linux x86_64; rv:154.0) Gecko/20100101 Firefox/154.0"
213.0.87.101 - - [27/Oct/2026:08:37:57 +0000] "GET / HTTP/1.1" 200 413 "-" "Mozilla/5.0 (X11; Linux x86_64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/156.0.0.0 Safari/537.36"
213.0.87.101 - - [27/Oct/2026:08:37:58 +0000] "GET /favicon.ico HTTP/1.1" 404 181 "http://98.91.198.128/" "Mozilla/5.0 (X11; Linux x86_64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/156.0.0.0 Safari/537.36"
34.122.147.229 - - [27/Oct/2026:08:38:13 +0000] "GET / HTTP/1.1" 200 413 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/125.0.6422.60 Safari/537.36 Edge/12.246"
admin@ip-172-31-34-242:~$
```

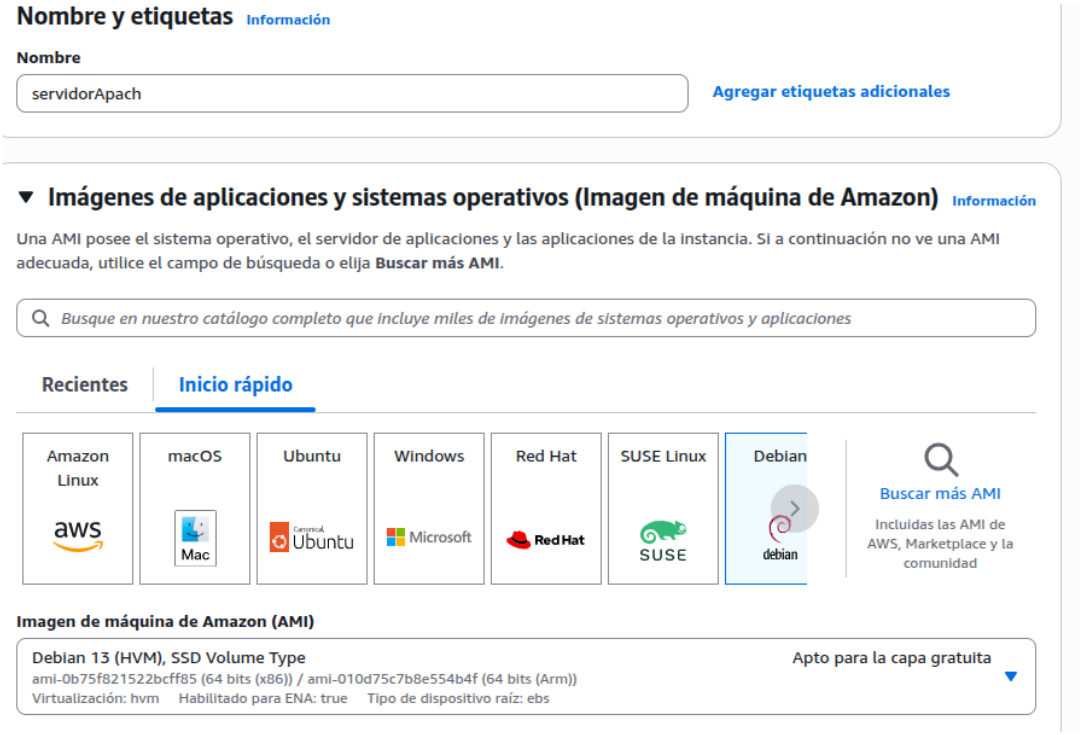
- error.log

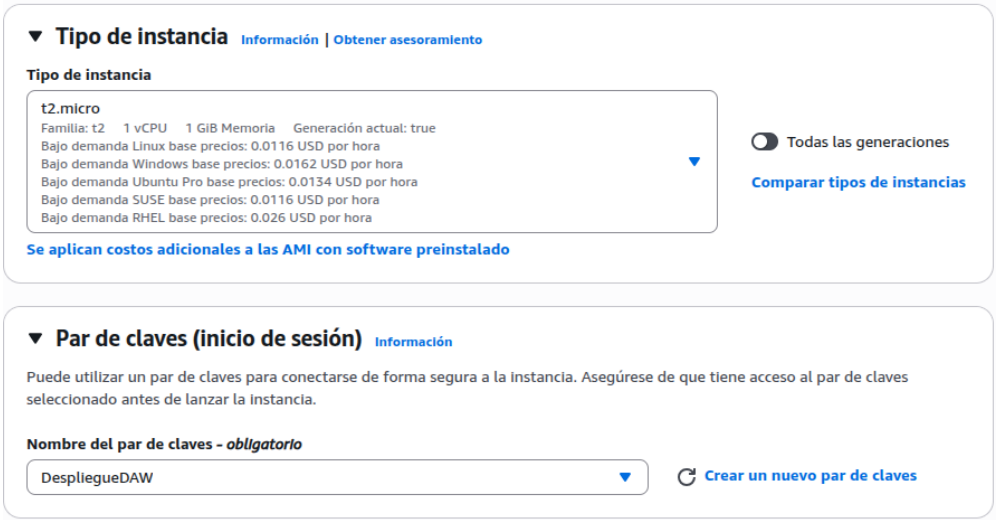
```
admin@ip-172-31-34-242:~$ cat /var/log/nginx/error.log
2026/10/26 12:38:20 [notice] 15931#15931: using inherited sockets from "5;6;"
```

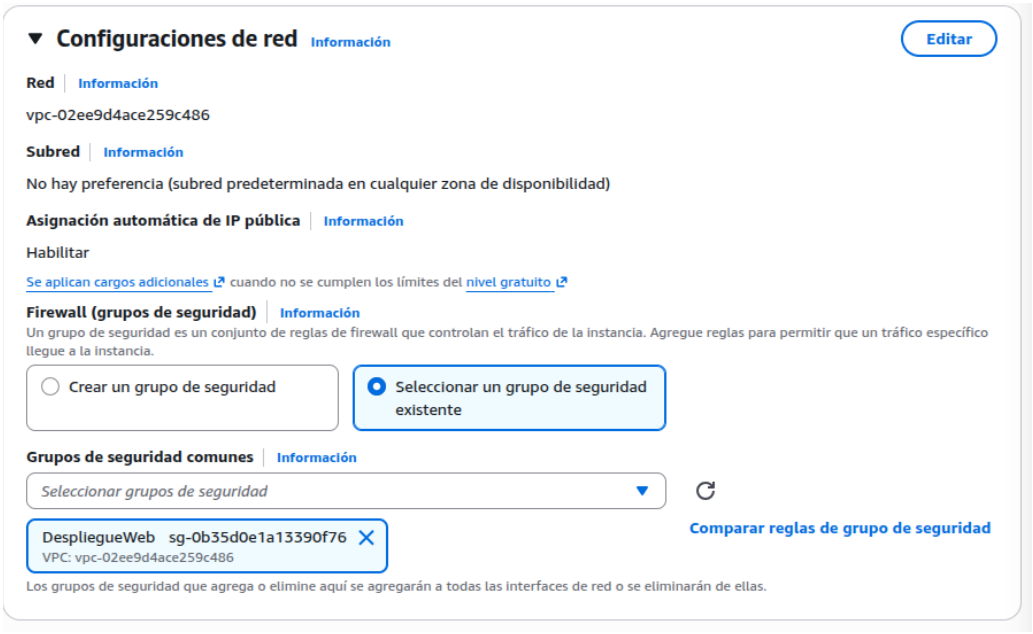
# Apache AWS

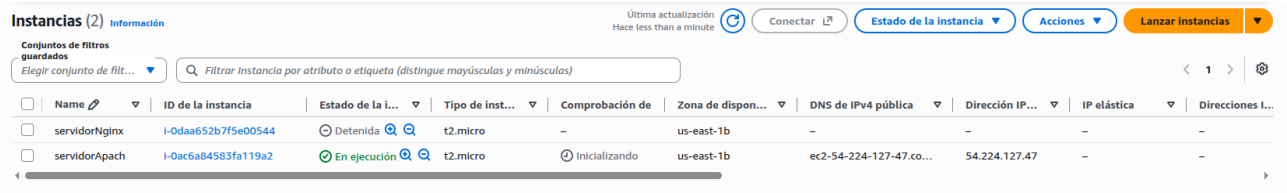
Se va a crear una instancia en AWS con estas caracteristicas:

- Nombre: servidorApach
- SO: Debian
- Tipo de instancia: t2.micro
- Par de clave: DespliegueDAW
- Grupo de seguridad: DespliegueWeb






Nos conectamos por SSH y actualizamos los repositorios y paquetes ya instalados

```
ivan@ivan-kubuntu ~/Documentos/2º DAW/Despliege aplicaiones Web
❯ ssh -i "DespliegueDAW.pem" admin@ec2-54-224-127-47.compute-1.amazonaws.com
ksshaskpass: Unable to parse phrase "The authenticity of host 'ec2-54-224-127-47.compute-1.amazonaws.com (54.224.127
.47)' can't be established.\nED25519 key fingerprint is: SHA256:g6J4Sx9zY+jKYSkkY3JoLCOi0fO12l5peibBwoJ2Gz8\nThis ke
y is not known by any other names.\nAre you sure you want to continue connecting (yes/no/[fingerprint])? "
Warning: Permanently added 'ec2-54-224-127-47.compute-1.amazonaws.com' (ED25519) to the list of known hosts.
Linux ip-172-31-47-118 6.12.74+deb13+1-cloud-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.12.74-2 (2026-03-08) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
admin@ip-172-31-47-118:~$ sudo apt update && sudo apt upgrade
Get:1 file:/etc/apt/mirrors/debian.list Mirrorlist [38 B]
Get:2 file:/etc/apt/mirrors/debian-security.list Mirrorlist [47 B]
Get:3 https://cdn-aws.deb.debian.org/debian trixie InRelease [140 kB]
Get:4 https://cdn-aws.deb.debian.org/debian trixie-updates InRelease [47.3 kB]
Get:5 https://cdn-aws.deb.debian.org/debian trixie-backports InRelease [54.0 kB]
Get:6 https://cdn-aws.deb.debian.org/debian-security trixie-security InRelease [43.4 kB]
Get:7 https://cdn-aws.deb.debian.org/debian trixie/main Sources [10.5 MB]
Get:8 https://cdn-aws.deb.debian.org/debian trixie/main amd64 Packages [9678 kB]
Get:9 https://cdn-aws.deb.debian.org/debian trixie/main Translation-en [6486 kB]
Get:10 https://cdn-aws.deb.debian.org/debian trixie-updates/main Sources [820 B]
Get:11 https://cdn-aws.deb.debian.org/debian trixie-updates/main amd64 Packages [720 B]
Get:12 https://cdn-aws.deb.debian.org/debian trixie-updates/main Translation-en [772 B]
Get:13 https://cdn-aws.deb.debian.org/debian trixie-backports/main Sources [335 kB]
Get:14 https://cdn-aws.deb.debian.org/debian trixie-backports/main amd64 Packages [294 kB]
Get:15 https://cdn-aws.deb.debian.org/debian trixie-backports/main Translation-en [230 kB]
Get:16 https://cdn-aws.deb.debian.org/debian-security trixie-security/main Sources [241 kB]
Get:17 https://cdn-aws.deb.debian.org/debian-security trixie-security/main amd64 Packages [270 kB]
Get:18 https://cdn-aws.deb.debian.org/debian-security trixie-security/main Translation-en [166 kB]
90% [8 Packages store 0 B]
```

Se instala apache2 desde los repositorios oficiales

```
admin@ip-172-31-47-118:~$ sudo apt install apache2
Installing:
  apache2

Installing dependencies:
  apache2-bin     libapr1t64               libaprutil1t64      liblua5.4-0  perl-modules-5.40
  apache2-data    libaprutil1-dbd-sqlite3  libgdbm-compat4t64  libperl5.40  ssl-cert
  apache2-utils   libaprutil1-ldap         libjansson4         perl

Suggested packages:
  apache2-doc              ufw          libterm-readline-gnu-perl     libtap-harness-archive-perl
  apache2-suexec-pristine  www-browser  | libterm-readline-perl-perl
  | apache2-suexec-custom  perl-doc     make

Summary:
  Upgrading: 0, Installing: 15, Removing: 0, Not Upgrading: 0
  Download size: 10.1 MB
  Space needed: 60.1 MB / 6621 MB available

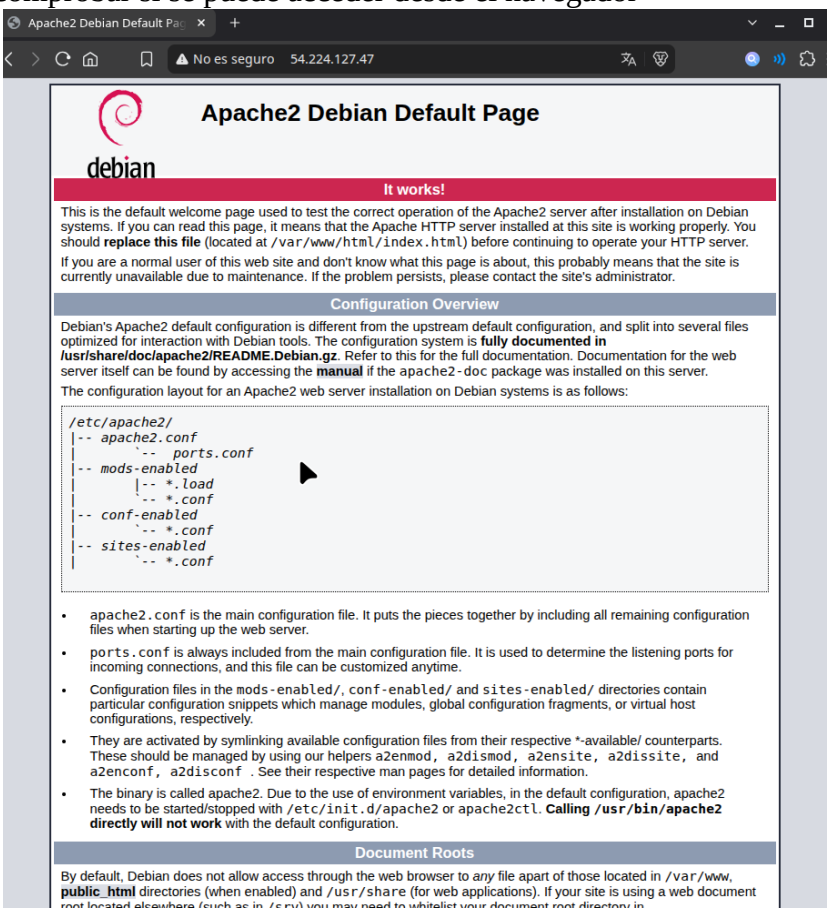
Continue? [Y/n]
```

A continuación se comprueba que el servicio de apache esta corriendo

```
admin@ip-172-31-47-118:~$ systemctl status apache2
● apache2.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
     Active: active (running) since Tue 2026-10-27 08:56:48 UTC; 1min 23s ago
 Invocation: 8cfd5461ccc94db6b49d8abd1f75efab
       Docs: https://httpd.apache.org/docs/2.4/
   Main PID: 16260 (apache2)
      Tasks: 55 (limit: 1127)
     Memory: 4.8M (peak: 5M)
        CPU: 34ms
     CGroup: /system.slice/apache2.service
             ├─16260 /usr/sbin/apache2 -k start
             ├─16261 /usr/sbin/apache2 -k start
             └─16282 /usr/sbin/apache2 -k start

Oct 27 08:56:48 ip-172-31-47-118 systemd[1]: Starting apache2.service - The Apache HTTP Server...
Oct 27 08:56:48 ip-172-31-47-118 apachectl[16259]: AH00558: apache2: Could not reliably determine the server's full>
Oct 27 08:56:48 ip-172-31-47-118 systemd[1]: Started apache2.service - The Apache HTTP Server.
lines 1-17/17 (END)
```

Como se ha reutilizado el grupo de seguridad no es necesario repetir los pasos y pasamos directamente a comprobar si se puede acceder desde el navegador



Accedemos al archivo HTML de apache y lo modificamos para personalizarlo

```
admin@ip-172-31-47-118:~$ sudo nano /var/www/html/index.html
```

```
 GNU nano 8.4                                /var/www/html/index.html *
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN" "http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional
<html xmlns="http://www.w3.org/1999/xhtml">
  <head>
    <meta http-equiv="Content-Type" content="text/html; charset=UTF-8" />
    <title>Ivan - Apache2 Debian Default Page: It works</title>
    <style type="text/css" media="screen">
  * {
    margin: 0px 0px 0px 0px;
    padding: 0px 0px 0px 0px;
```

```
 GNU nano 8.4                                /var/www/html/index.html *
  }
    </style>
  </head>
  <body>
    <div class="main_page">
      <div class="page_header floating_element">
        <img src="/icons/openlogo-75.png" alt="Debian Logo" class="floating_element"/>
        <span class="floating_element">
          Bienvenido al Apache2 de Ivan
        </span>
      </div>
<!--      <div class="table_of_contents floating_element">
```

Volvemos al nevegador y actualizamos viendo como el cambio que se ha hecho en el HTML se ha aplicado en la pagina web




Ahora se revisan los logs de acceso y los logs de error

- access.log

```
admin@ip-172-31-47-118:~$ cat /var/log/apache2/access.log
213.0.87.101 - - [27/Oct/2026:09:00:52 +0000] "GET / HTTP/1.1" 200 3383 "-" "Mozilla/5.0 (X11; Linux x86_64) AppleWe
bKit/537.36 (KHTML, like Gecko) Chrome/156.0.0.0 Safari/537.36"
213.0.87.101 - - [27/Oct/2026:09:00:53 +0000] "GET /icons/openlogo-75.png HTTP/1.1" 200 6040 "http://54.224.127.47/"
 "Mozilla/5.0 (X11; Linux x86_64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/156.0.0.0 Safari/537.36"
213.0.87.101 - - [27/Oct/2026:09:00:54 +0000] "GET /favicon.ico HTTP/1.1" 404 531 "http://54.224.127.47/" "Mozilla/5
.0 (X11; Linux x86_64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/156.0.0.0 Safari/537.36"
34.122.147.229 - - [27/Oct/2026:09:01:00 +0000] "GET / HTTP/1.1" 200 3383 "-" "Mozilla/5.0 (Windows NT 10.0; Win64;
x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/125.0.6422.60 Safari/537.36 Edge/12.246"
34.122.147.229 - - [27/Oct/2026:09:01:02 +0000] "GET /icons/openlogo-75.png HTTP/1.1" 200 6040 "http://54.224.127.47
/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/125.0.6422.60 Safari/537
.36 Edge/12.246"
45.156.87.44 - - [27/Oct/2026:09:04:05 +0000] "GET / HTTP/1.0" 200 10977 "-" "Mozilla/5.0 (compatible; opendir-scann
er/1.0)"
45.156.87.44 - - [27/Oct/2026:09:04:05 +0000] "GET / HTTP/1.1" 200 3346 "-" "Mozilla/5.0 (compatible; opendir-scanne
r/1.0)"
213.0.87.101 - - [27/Oct/2026:09:09:56 +0000] "GET / HTTP/1.1" 200 3403 "-" "Mozilla/5.0 (X11; Linux x86_64) AppleWe
bKit/537.36 (KHTML, like Gecko) Chrome/156.0.0.0 Safari/537.36"
20.150.210.168 - - [27/Oct/2026:09:12:10 +0000] "GET /developmentserver/metadatauploader HTTP/1.1" 404 476 "-" "Mozi
lla/5.0 zgrab/0.x"
admin@ip-172-31-47-118:~$
```

- error.log

```
admin@ip-172-31-47-118:~$ cat /var/log/apache2/error.log
[Tue Oct 27 08:56:48.717634 2026] [mpm_event:notice] [pid 16260:tid 16260] AH00489: Apache/2.4.69 (Debian) configure
d -- resuming normal operations
[Tue Oct 27 08:56:48.717688 2026] [core:notice] [pid 16260:tid 16260] AH00094: Command line: '/usr/sbin/apache2'
admin@ip-172-31-47-118:~$
```

# Nginx dockerizado Dockerfile

Se crea el Dockerfile para crear la imagen con nginx con la imagen base debian [en el paso a paso se dice que hay que usar una imagen base de Ubuntu, luego dice alpine (el cual no se puede usar apt ya que usa apk) y en el ejemplo se usa Debian. Como se ha usado Debian en AWS he decidido mantener la imagen base Debian].

```
ivan@ivan-kubuntu ~/Documentos/2º DAW/Despliege aplicaiones Web/apuntes-daw-2627/docker/servidorWeb/nginx  main
❯ nano Dockerfile
```

```
  GNU nano 8.7.1                                        Dockerfile
FROM debian:latest
MAINTAINER Ivan Rubert <ivanrubert23@gmail.com>

# Actualizamos e instalamos Nginx
RUN apt update && apt upgrade -y && apt install -y nginx && \
    rm -rf /var/lib/apt/lists/*

# Exponemos el puerto 80
EXPOSE 80

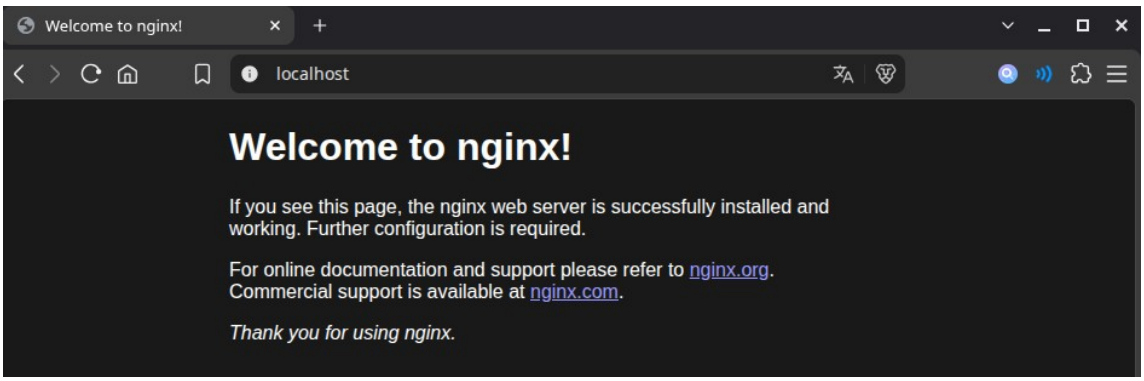
# Comando para iniciar Nginx en primer plano
CMD ["nginx", "-g", "daemon off;"]
```

Se crea la imagen con docker build y se lanza el contenedor con docker run

```
ivan@ivan-kubuntu ~/Documentos/2º DAW/Despliege aplicaiones Web/apuntes-daw-2627/docker/servidorWeb/nginx  main
❯ docker build -t servernginx .
[+] Building 4.2s (3/5)                                                                          docker:default
 => [internal] load build definition from Dockerfile                                                      0.0s
 => => transferring dockerfile: 352B                                                                      0.0s
 => WARN: MaintainerDeprecated: Maintainer instruction is deprecated in favor of using label (line 2)     0.0s
 => [internal] load metadata for docker.io/library/debian:latest                                          1.5s
 => [internal] load .dockerignore                                                                         0.0s
 => => transferring context: 2B                                                                           0.0s
 => [1/2] FROM docker.io/library/debian:latest@sha256:4c25275446ca371766b8b093c696782eabce06263fccacb8d5d813b  2.6s
 => => resolve docker.io/library/debian:latest@sha256:4c25275446ca371766b8b093c696782eabce06263fccacb8d5d813b  0.0s
 => => sha256:8d0037029946d2c6dabce18eceda79c533c5797ed1ee86f94084cb8cddaeea91 49.39MB / 49.39MB           2.1s
 => => extracting sha256:8d0037029946d2c6dabce18eceda79c533c5797ed1ee86f94084cb8cddaeea91                  0.5s
```

```
ivan@ivan-kubuntu ~/Documentos/2º DAW/Despliege aplicaiones Web/apuntes-daw-2627/docker/servidorWeb/nginx  main
❯ docker run --name=miservernginx -d -p 80:80 servernginx
4a9f623ea80e296c65ab6a93a80419c19dbf8837ffeb9ecebfc81d7439577a95
```

Se comprueba que el servicio de nginx esta corriendo buscando http://localhost en el navegador



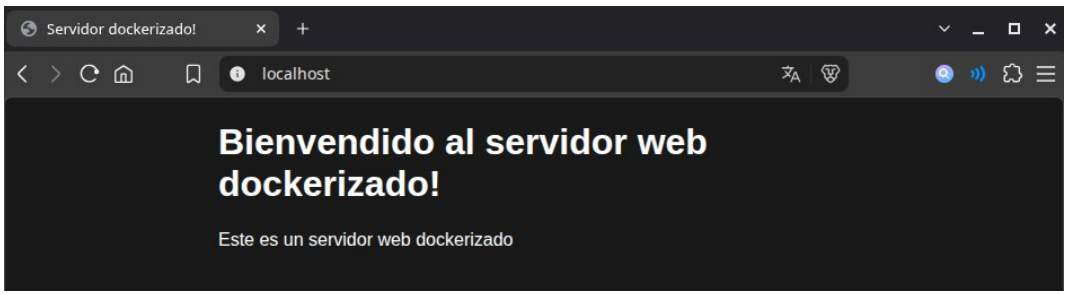

A continuación eliminamos el contenedor y lo volvemos a lanzar con un bind mount con un html personalizado (en la practica usa ~/web pero en mi caso la ruta cambia a ~/Documentos/2º\ DAW/Despliege\ aplicaiones\ Web/apuntes-daw-2627/docker/servidorWeb/nginx/web para asi tener organizadas las practicas)

```
ivan@ivan-kubuntu ~/Documentos/2º DAW/Despliege aplicaiones Web/apuntes-daw-2627/docker/servidorWeb/nginx  main
❯ docker stop miservernginx && docker rm miservernginx
miservernginx
miservernginx
```

```
ivan@ivan-kubuntu ~/Documentos/2º DAW/Despliege aplicaiones Web/apuntes-daw-2627/docker/servidorWeb/nginx  main
❯ mkdir web
ivan@ivan-kubuntu ~/Documentos/2º DAW/Despliege aplicaiones Web/apuntes-daw-2627/docker/servidorWeb/nginx  main
❯ nano web/index.html
```

```
 GNU nano 8.7.1                                    web/index.html *
<!DOCTYPE html>
<html>
<head>
<title>Servidor dockerizado!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Bienvendido al servidor web dockerizado!</h1>
<p> Este es un servidor web dockerizado </p>
</body>
</html>
```

```
❯ docker run -d --name=miservernginx \
    --mount type=bind,source=/home/ivan/Documentos/2º\ DAW/Despliege\ aplicaiones\ Web/apuntes-daw-2627/docker/servidorWeb/nginx/web,target=/var/www/html \
    -p 80:80 \
    servernginx
```

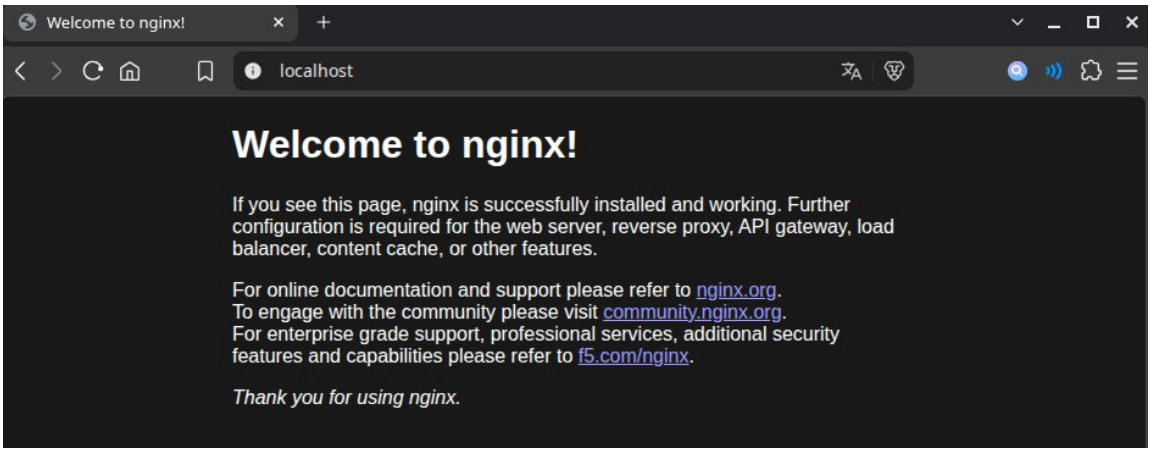



# Nginx dockerizado imagen oficiales

Para poder usar el mismo puerto se debe parar y eliminar el contenedor anterior y se vuelve a lanzar con la imagen oficial de nginx

```
ivan@ivan-kubuntu ~/Documentos/2º DAW/Despliege aplicaiones Web/apuntes-daw-2627/docker/servidorWeb/nginx  main
❯ docker stop miservernginx && docker rm miservernginx
miservernginx
miservernginx
```

```
ivan@ivan-kubuntu ~/Documentos/2º DAW/Despliege aplicaiones Web/apuntes-daw-2627/docker/servidorWeb/nginx  main
❯ docker run -d --name=miservernginx2 \
    --mount type=bind,source=/home/ivan/Documentos/2º\ DAW/Despliege\ aplicaiones\ Web/apuntes-daw-2627/docker/servi
dorWeb/nginx/web,target=/var/www/html \
    -p 80:80 \
    nginx:latest
Unable to find image 'nginx:latest' locally
latest: Pulling from library/nginx
8334177aa35b: Download complete
9a2c3eb3312e: Download complete
dd00b11a22aa: Download complete
d5f3a4724daf: Extracting 1 s
24bf03b6a474: Download complete
f2be67225471: Pull complete
a856bfc2a772: Download complete
ee284e8d8c11: Download complete
6542b5f86ffd: Download complete
```

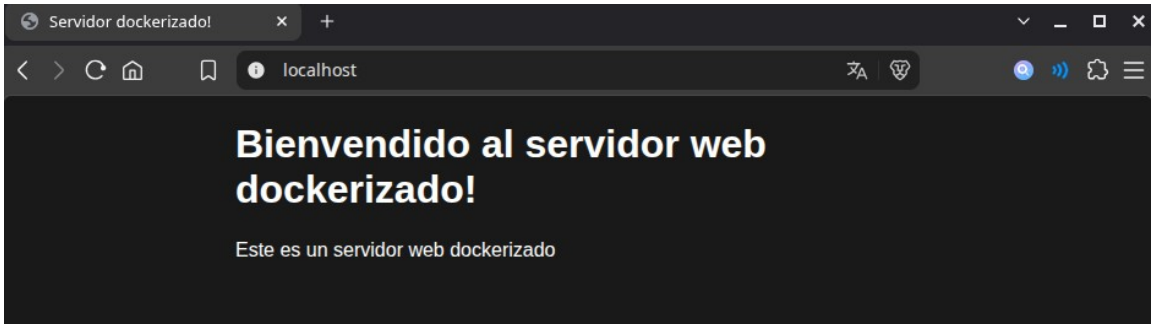



No sale el html que teniamos debido a que la imagen oficial de nginx usa la ruta /usr/share/nginx/html asi que se vuelve a eliminar el contenedor y lo volvemos a lanzar correctamente. Ademas se ha escogido la opción de :ro (read only) para que el contenedor solo pueda leer la información de la carpeta montada y los cambios se realicen en la carpeta host

```
ivan@ivan-kubuntu ~/Documentos/2º DAW/Despliege aplicaiones Web/apuntes-daw-2627/docker/servidorWeb/nginx  main
❯ docker stop miservernginx2 && docker rm miservernginx2
miservernginx2
miservernginx2
```

```
ivan@ivan-kubuntu ~/Documentos/2º DAW/Despliege aplicaiones Web/apuntes-daw-2627/docker/servidorWeb/nginx  main
❯ docker run -d --name=miservernginx2 \
    -v /home/ivan/Documentos/2º\ DAW/Despliege\ aplicaiones\ Web/apuntes-daw-2627/docker/servidorWeb/nginx/web:/usr/
share/nginx/html:ro \
    -p 80:80 \
    nginx:latest
0a6f99f30a6dd2ff6a0536f34ee73521732a710c7243691524eac1dbd047f502
```

Despues de cambiar la ruta donde se coloca el html revisamos el navegador y vemos que ya ha cambiado




# Apache2 dockerizado con Dockerfile

Para empezar a hacer este ejercicio usando el mismo puerto primero se debe parar y eliminar el contenedor anterior

```
ivan@ivan-kubuntu ~/Documentos/2º DAW/Despliege aplicaiones Web/apuntes-daw-2627/docker/servidorWeb/apache  main
❯ docker stop miservernginx2 && docker rm miservernginx2
miservernginx2
miservernginx2
```

Ahora que ya no están los contenedores y los puertos están libres empezamos creando un Dockerfile

```
ivan@ivan-kubuntu ~/Documentos/2º DAW/Despliege aplicaiones Web/apuntes-daw-2627/docker/servidorWeb/apache  main
❯ nano Dockerfile
```

```
  GNU nano 8.7.1                                  Dockerfile *
FROM debian:latest
MAINTAINER Ivan Rubert <ivanrubert23@gmail.com>

# Actualizamos e instalamos Apache2
RUN apt update && apt upgrade -y && apt install -y apache2 && \
    rm -rf /var/lib/apt/lists/*

# Exponemos el puerto 80
EXPOSE 80

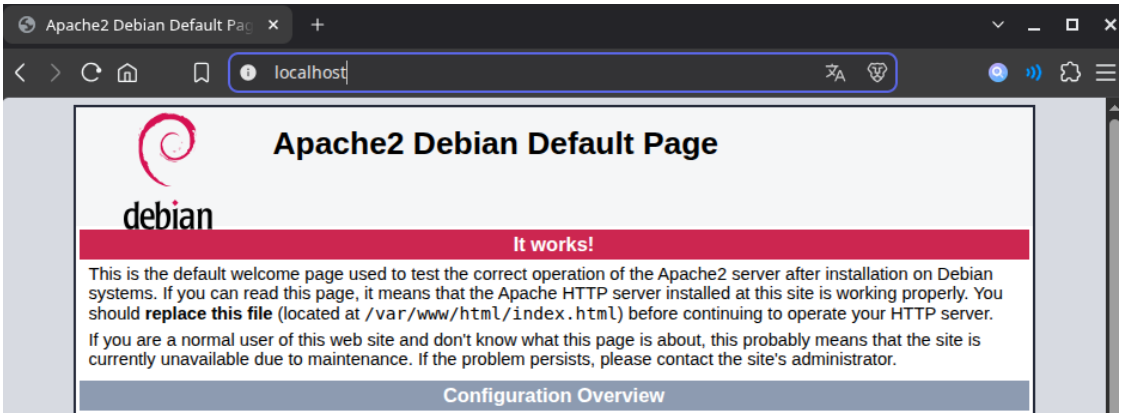
# Comando para iniciar Apache2 en primer plano
CMD ["apache2ctl", "-D", "FOREGROUND"]
```

Una vez ya creado el Dockerfile se crea la imagen con docker build y se despliega el contenedor con esta imagen creada

```
ivan@ivan-kubuntu ~/Documentos/2º DAW/Despliege aplicaiones Web/apuntes-daw-2627/docker/servidorWeb/apache  main
❯ docker build -t serverapache .
[+] Building 6.0s (4/5)                                                                                 docker:default
 => [internal] load build definition from Dockerfile                                                             0.0s
 => => transferring dockerfile: 358B                                                                             0.0s
 => WARN: MaintainerDeprecated: Maintainer instruction is deprecated in favor of using label (line 2)            0.0s
 => [internal] load metadata for docker.io/library/debian:latest                                                 1.0s
 => [internal] load .dockerignore                                                                                0.0s
 => => transferring context: 2B                                                                                  0.0s
 => CACHED [1/2] FROM docker.io/library/debian:latest@sha256:4c25275446ca371766b8b093c696782eabce06263fccacb8    0.0s
 => => resolve docker.io/library/debian:latest@sha256:4c25275446ca371766b8b093c696782eabce06263fccacb8d5d813b    0.0s
 => [2/2] RUN apt update && apt upgrade -y && apt install -y apache2 &&     rm -rf /var/lib/apt/lists/*         4.9s
 => => # gnu/perl/5.40 /usr/share/perl/5.40 /usr/local/lib/site_perl) at /usr/share/perl5/Debconf/FrontEnd/Readline
 => => # .pm line 8, <STDIN> line 1.)
 => => # debconf: falling back to frontend: Teletype
 => => # debconf: unable to initialize frontend: Teletype
 => => # debconf: (This frontend requires a controlling tty.)
 => => # debconf: falling back to frontend: Noninteractive
```

```
ivan@ivan-kubuntu ~/Documentos/2º DAW/Despliege aplicaiones Web/apuntes-daw-2627/docker/servidorWeb/apache  main
❯ docker run --name=miserverapache -d -p 80:80 serverapache:latest
e149404d23be575cd3e58ad3d0a423b2e4b675416067e5982c8ff4f61307bbc3
```

Una vez desplegado revisamos el navegador la pagina web




Una vez comprobado continuamos eliminando el contenedor para poner un html personalizado a través de bind mount (en lugar de usar *~/web* se usara para tener organizadas las practicas *~/Documentos/2º\ DAW/Despliege\ aplicaiones\ Web/apuntes-daw-2627/docker/servidorWeb/apache/web*)

```
ivan@ivan-kubuntu ~/Documentos/2º DAW/Despliege aplicaiones Web/apuntes-daw-2627/docker/servidorWeb/apache  main
❯ docker stop miserverapache && docker rm miserverapache
miserverapache
miserverapache
```

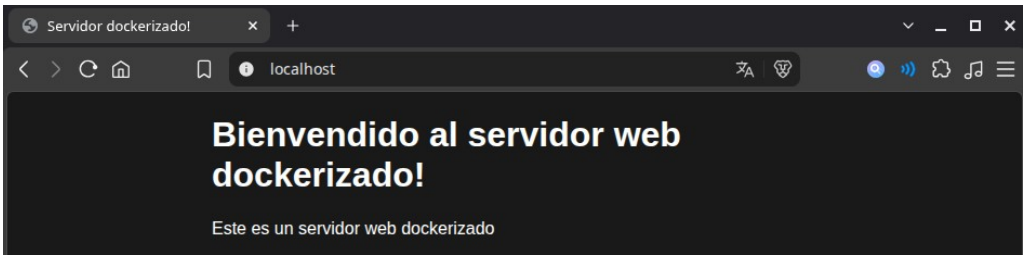
```
ivan@ivan-kubuntu ~/Documentos/2º DAW/Despliege aplicaiones Web/apuntes-daw-2627/docker/servidorWeb/apache  main
❯ mkdir web
ivan@ivan-kubuntu ~/Documentos/2º DAW/Despliege aplicaiones Web/apuntes-daw-2627/docker/servidorWeb/apache  main
❯ nano web/index.html
```

```
 GNU nano 8.7.1                                        web/index.html *
<!DOCTYPE html>
<html>
<head>
<title>Servidor dockerizado!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Bienvendido al servidor web dockerizado!</h1>
<p> Este es un servidor web dockerizado </p>
</body>
</html>
```

Una vez preparado el html se despliega el contenedor con el bind mount

```
ivan@ivan-kubuntu ~/Documentos/2º DAW/Despliege aplicaiones Web/apuntes-daw-2627/docker/servidorWeb/apache  main
❯ docker run -d --name=miserverapache \
    --mount type=bind,source=/home/ivan/Documentos/2º\ DAW/Despliege\ aplicaiones\ Web/apuntes-daw-2627/docker/servi
dorWeb/apache/web,target=/var/www/html \
    -p 80:80 \
    serverapache:latest
```

Una vez desplegado revisamos en el navegador que se haya creado bien el bind mount y salga la pagina web




# Apache2 dockerizado con imagen oficial

Para hacer este ejercicio con el puerto 80 (el mismo que el anterior), primero se debe parar y eliminar el contenedor anterior

```
ivan@ivan-kubuntu ~/Documentos/2º DAW/Despliege aplicaiones Web/apuntes-daw-2627/docker/servidorWeb/apache  main
❯ docker stop miserverapache && docker rm miserverapache
miserverapache
miserverapache
```

Ahora que ya tenemos el puerto libre lanzamos el contenedor con la imagen oficial que usa la ruta */usr/local/apache2/htdocs*

```
ivan@ivan-kubuntu ~/Documentos/2º DAW/Despliege aplicaiones Web/apuntes-daw-2627/docker/servidorWeb/apache  main
❯ docker run -d --name=miserverapache2 \
    -v /home/ivan/Documentos/2º\ DAW/Despliege\ aplicaiones\ Web/apuntes-daw-2627/docker/servidorWeb/apache/web:/usr
/local/apache2/htdocs:ro \
    -p 80:80 \
    httpd:latest
Unable to find image 'httpd:latest' locally
latest: Pulling from library/httpd
b361b1ecdecd: Pull complete
f8903be41a4f: Pull complete
720ea190dd65: Pull complete
f2be67225471: Already exists
4f4fb700ef54: Pull complete
1854ebd7c33a: Pull complete
8d61ccc1b972: Download complete
096bb8816822: Download complete
Digest: sha256:a30cd4cd57f3f3c3be9b2d16384d2488839996bb14c67df2b51f1ecbb3e584a1
Status: Downloaded newer image for httpd:latest
a14ec0af612552e02a98234a9db75848f888fbac95278d5ed4027e281c89b8aa
```

Y se revisa en el navegador que la pagina web se muestra con el index.html modificado

